Facial Nerve

ENT | 3rd Year MBBS

Hi Sammy!
CN VII = most tested cranial nerve in ENT!
Master the anatomy!

Facial Nerve | Bell's Palsy | CN VII | ENT

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Facial Nerve Anatomy + Course

CN VII has motor, sensory & parasympathetic fibers! Remember all 3!

Course of Facial Nerve (CN VII)

Segment	Key Details
Intracranial	Arises from pons (motor nucleus + nervus intermedius)
Meatal (IAM)	Travels with CN VIII through internal auditory meatus
Labyrinthine	Narrowest segment -> most vulnerable to compression
Geniculate ganglion	First genu; gives off greater petrosal nerve (tears)
Tympanic	Runs along medial wall of middle ear above oval window
Mastoid	Second genu; gives off nerve to stapedius & chorda tympani
Extracranial	Exits via stylomastoid foramen -> parotid gland -> 5 branches

Three Functional Components

Component	Function	Branch / Structure
Motor	Muscles of facial expression	Temporal, zygomatic, buccal, mand., cervical
Sensory	Taste — anterior 2/3 tongue	Chorda tympani nerve
Parasympathetic	Lacrimal, submandibular glands	Greater petrosal n. & chorda tympani

Five Terminal Branches of Facial Nerve

Temporal:	Frontalis, orbicularis oculi (upper)
Zygomatic:	Orbicularis oculi (lower), nasalis
Buccal:	Buccinator, orbicularis oris (upper)
Marginal mandibular:	Lower lip muscles (most important to preserve in surgery!)
Cervical:	Platysma

★ Mnemonic — 5 Branches!

"To Zanzibar By Motor Car" = Temporal, Zygomatic, Buccal, Marginal mandibular, Cervical

Labyrinthine segment = NARROWEST = most susceptible to swelling & compression!

UMN vs LMN Palsy — The Table

Forehead sparing = UMN! This one point = 5 marks guaranteed!

UMN vs LMN Facial Nerve Palsy

Feature	UMN (Upper Motor Neuron)	LMN (Lower Motor Neuron)
Forehead	SPARED (wrinkles forehead)	INVOLVED (cannot wrinkle)
Eye closure	Can close eye	Cannot close eye (lagoph.)
Affected side	Lower face contralateral	Entire face ipsilateral
Nasolabial fold	Flattened (lower face only)	Flattened (entire face)
Cause	Stroke, tumor, MS	Bell's, Ramsay Hunt, tumor
Emotional mvmt	May be affected	Usually preserved
Taste	Normal	May be lost (ant. 2/3)
Hyperacusis	Absent	Present (if N. to stapedius)

★ **Why is forehead SPARED in UMN lesions?**
Upper face (frontalis) receives BILATERAL cortical innervation. So even if one side is damaged, the other cortex still supplies it. Lower face = UNILATERAL supply.

Topographic Diagnosis — Level of Lesion

Level of Lesion	Features Present
Above N. to stapedius	LMN palsy + hyperacusis + loss of taste + dry eye
Above chorda tympani	LMN palsy + loss of taste + dry eye (no hyperacusis)
Above greater petrosal N.	LMN palsy + dry eye + loss of taste
At stylomastoid foramen	LMN palsy ONLY (no taste/tear/hyperacusis loss)
In parotid gland	Selected branch palsy only (e.g., marginal mandibular)

Common Causes of LMN Facial Palsy

Category	Causes
Idiopathic	Bell's palsy (most common cause — 72% of all facial palsies)
Infective	Ramsay Hunt (VZV), otitis media, mastoiditis, TB, HIV
Traumatic	Temporal bone fracture, birth trauma, surgical (parotid/mastoid)
Neoplastic	Parotid tumor, acoustic neuroma, cholesteatoma, glomus tumor
Others	Sarcoidosis, Guillain-Barre, Melkersson-Rosenthal syndrome

Bell's palsy = most common cause of facial palsy! Know it cold!

Bell's Palsy (Idiopathic Facial Palsy)

Definition:	Acute onset, idiopathic LMN facial palsy (diagnosis of exclusion)
Cause:	Possibly HSV-1 reactivation in geniculate ganglion
Onset:	Rapid (hours to 1-2 days), unilateral, often preceded by ear pain
Features:	Complete LMN palsy: cannot close eye, forehead, smile
Prognosis:	85% recover completely. Worse if complete paralysis, DM, elderly
Treatment:	Prednisolone (within 72 hrs) + eye care (artificial tears, taping)
Antiviral:	Acyclovir/valacyclovir — controversial, may add if severe

Ramsay Hunt Syndrome (Herpes Zoster Oticus)

Cause:	VZV reactivation in geniculate ganglion
Triad:	LMN facial palsy + vesicles on ear/EAC + otalgia
Vesicles:	On pinna, external auditory canal, tympanic membrane
May have:	SNHL, vertigo, tinnitus (CN VIII involvement)
Treatment:	Acyclovir/valacyclovir + prednisolone + eye care
Prognosis:	WORSE than Bell's palsy — only 50-60% full recovery

Bell's Palsy vs Ramsay Hunt Syndrome

Feature	Bell's Palsy	Ramsay Hunt Syndrome
Cause	HSV-1 (likely)	VZV (Herpes Zoster)
Vesicles	ABSENT	PRESENT (ear, EAC)
Hearing loss	Usually absent	SNHL may be present
Pain	Mild post-auricular	Severe otalgia (prodrome)
Prognosis	Good (85% recover)	Poorer (50-60% recover)
Treatment	Steroids +/- acyclovir	Acyclovir + steroids (must)

Bell's vs Hunt!
Bell's = no vesicles, better prognosis. Ramsay Hunt = VESICLES on ear + worse prognosis!

House-Brackmann grading
is asked in viva!
Know Grades I-VI!

House-Brackmann Facial Nerve Grading

Grade	Dysfunction	Description
I	Normal	Normal facial function in all areas
II	Mild	Slight weakness on close inspection, complete eye closure
III	Moderate	Obvious weakness, complete eye closure with effort, forehead mvmt
IV	Mod-Severe	Obvious weakness, incomplete eye closure, no forehead mvmt
V	Severe	Barely perceptible motion, incomplete eye closure
VI	Total	No movement at all (complete paralysis)

Investigations for Facial Nerve Palsy

Investigation	Purpose / Indication
Schirmer's test	Tear production — tests greater petrosal nerve function
Stapedial reflex	Tests nerve to stapedius — absent in proximal lesion
Taste testing	Electrogustometry — tests chorda tympani function
Nerve excitability test	Compares threshold on both sides (Hilger's)
Electroneuronography (ENoG)	Quantifies nerve degeneration (> 90% = poor prognosis)
EMG	Tests for denervation / reinnervation potential
MRI with Gd contrast	Rule out tumor, inflammation (enhancing nerve)
HRCT temporal bone	Fractures, cholesteatoma, mastoid disease

Management of Facial Nerve Palsy

Eye care:	Artificial tears, taping eyelid at night, moisture chamber
Medical:	Prednisolone 1mg/kg x 5 days, taper over 5 days (within 72 hrs)
Antiviral:	Acyclovir 400mg 5x/day x 7-10 days (if Ramsay Hunt / severe)
Physiotherapy:	Facial exercises, electrical stimulation, massage
Surgery:	Facial nerve decompression if > 90% degeneration on ENoG

★ Eye Care = Priority!
Most important complication of facial palsy = EXPOSURE KERATITIS (corneal damage).
Always protect the eye! Artificial tears by day, tape shut at night. Refer to ophthal if Grade IV+.

Facial nerve = 15+
marks in ENT exam!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Describe anatomy of facial nerve with its course & branches
- 2. Differentiate UMN vs LMN facial palsy with causes
- 3. Describe Bell's palsy — features, investigations & management
- 4. Describe Ramsay Hunt syndrome & compare with Bell's palsy
- 5. House-Brackmann grading of facial nerve palsy with management

Most Repeated SAQs (5–6 marks)

- 1. Forehead sparing in UMN lesion — explain why
- 2. Topographic diagnosis of facial nerve palsy
- 3. Schirmer's test — technique & significance
- 4. Five branches of facial nerve with muscles supplied
- 5. Eye care in facial nerve palsy
- 6. Melkersson-Rosenthal syndrome

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common cause of facial palsy	Bell's palsy (72%)
Forehead spared in facial palsy	UMN lesion
Vesicles on ear + facial palsy	Ramsay Hunt syndrome (VZV)
Narrowest segment of facial canal	Labyrinthine segment
Nerve to stapedius is damaged	Hyperacusis present
Chorda tympani is damaged	Loss of taste anterior 2/3 tongue
Crocodile tears (Bogorad syndrome)	Aberrant regeneration of CN VII
Bell's phenomenon	Upward roll of eye on attempted closure
Melkersson-Rosenthal syndrome	Recurrent facial palsy + lip edema + fissured tongue
ENoG > 90% degeneration	Consider facial nerve decompression
Facial nerve exits skull through	Stylomastoid foramen
5 branches mnemonic	To Zanzibar By Motor Car

♥ **A smile is the most beautiful expression, Sammy!**
You'll help restore smiles one day. Keep learning, future doctor!